DefineAndSolveMLProblem

July 30, 2025

1 Lab 8: Define and Solve an ML Problem of Your Choosing

[1]:
```
import pandas as pd
import numpy as np
import os
import matplotlib.pyplot as plt
import seaborn as sns
```

In this lab assignment, you will follow the machine learning life cycle and implement a model to solve a machine learning problem of your choosing. You will select a data set and choose a predictive problem that the data set supports. You will then inspect the data with your problem in mind and begin to formulate a project plan. You will then implement the machine learning project plan.

You will complete the following tasks:

1. Build Your DataFrame
2. Define Your ML Problem
3. Perform exploratory data analysis to understand your data.
4. Define Your Project Plan
5. Implement Your Project Plan:
    - Prepare your data for your model.
    - Fit your model to the training data and evaluate your model.
    - Improve your model's performance.

1.1 Part 1: Build Your DataFrame

You will have the option to choose one of four data sets that you have worked with in this program:

- The "census" data set that contains Census information from 1994: `censusData.csv`
- Airbnb NYC "listings" data set: `airbnbListingsData.csv`
- World Happiness Report (WHR) data set: `WHR2018Chapter2OnlineData.csv`
- Book Review data set: `bookReviewsData.csv`

Note that these are variations of the data sets that you have worked with in this program. For example, some do not include some of the preprocessing necessary for specific models.

Load a Data Set and Save it as a Pandas DataFrame The code cell below contains filenames (path + filename) for each of the four data sets available to you.

Task: In the code cell below, use the same method you have been using to load the data using `pd.read_csv()` and save it to DataFrame `df`.

You can load each file as a new DataFrame to inspect the data before choosing your data set.

[3]:
```
# File names of the four data sets
adultDataSet_filename = os.path.join(os.getcwd(), "data", "censusData.csv")
airbnbDataSet_filename = os.path.join(os.getcwd(), "data", "airbnbListingsData.
 ↪csv")
WHRDataSet_filename = os.path.join(os.getcwd(), "data",␣
 ↪"WHR2018Chapter2OnlineData.csv")
bookReviewDataSet_filename = os.path.join(os.getcwd(), "data", "bookReviewsData.
 ↪csv")


df = pd.read_csv(bookReviewDataSet_filename)# YOUR CODE HERE
    # I selected the book review dataset
df.head()
```

[3]:

	Review	Positive Review
0	This was perhaps the best of Johannes Steinhof…	True
1	This very fascinating book is a story written …	True
2	The four tales in this collection are beautifu…	True
3	The book contained more profanity than I expec…	False
4	We have now entered a second time of deep conc…	True

1.2 Part 2: Define Your ML Problem

Next you will formulate your ML Problem. In the markdown cell below, answer the following questions:

1. List the data set you have chosen.
2. What will you be predicting? What is the label?
3. Is this a supervised or unsupervised learning problem? Is this a clustering, classification or regression problem? Is it a binary classificaiton or multi-class classifiction problem?
4. What are your features? (note: this list may change after your explore your data)
5. Explain why this is an important problem. In other words, how would a company create value with a model that predicts this label?

1. I selected the book review dataset.
2. I would like to predict whether a book review is positive or negative based on the 'Positive Review' column label.
3. This is a supervised learning problem because the label is provided and would be categorized under classification. More specifically, the classification problem is a binary type because there are only two possible outcomes: (1) positive or (0) negative.
4. The feature of this book review dataset is the 'Review' column, which provides written details to support the outcomes of the 'positive review' label.

5. This is an important problem to solve as it can support book recommendations and categorize books effectively by genre and/or topic. Companies such as Amazon or Barnes and Noble can gain value from these predictions to enhance customer activity and engagement on their online platforms.

1.3 Part 3: Understand Your Data

The next step is to perform exploratory data analysis. Inspect and analyze your data set with your machine learning problem in mind. Consider the following as you inspect your data:

1. What data preparation techniques would you like to use? These data preparation techniques may include:
    - addressing missingness, such as replacing missing values with means
    - finding and replacing outliers
    - renaming features and labels
    - finding and replacing outliers
    - performing feature engineering techniques such as one-hot encoding on categorical features
    - selecting appropriate features and removing irrelevant features
    - performing specific data cleaning and preprocessing techniques for an NLP problem
    - addressing class imbalance in your data sample to promote fair AI
2. What machine learning model (or models) you would like to use that is suitable for your predictive problem and data?
    - Are there other data preparation techniques that you will need to apply to build a balanced modeling data set for your problem and model? For example, will you need to scale your data?
3. How will you evaluate and improve the model's performance?
    - Are there specific evaluation metrics and methods that are appropriate for your model?

Think of the different techniques you have used to inspect and analyze your data in this course. These include using Pandas to apply data filters, using the Pandas `describe()` method to get insight into key statistics for each column, using the Pandas `dtypes` property to inspect the data type of each column, and using Matplotlib and Seaborn to detect outliers and visualize relationships between features and labels. If you are working on a classification problem, use techniques you have learned to determine if there is class imbalance.

Task: Use the techniques you have learned in this course to inspect and analyze your data. You can import additional packages that you have used in this course that you will need to perform this task.

Note: You can add code cells if needed by going to the Insert menu and clicking on Insert Cell Below in the drop-drown menu.

[4]:
```
# YOUR CODE HERE

# Part 3: Understand Your Data
```

```python
# Display basic structure
print("Data Types:\n", df.dtypes)
print("\nMissing Values:\n", df.isnull().sum())

# Check class balance
print("\nClass Distribution:\n", df['Positive Review'].value_counts())

# Visualize class distribution
sns.countplot(data=df, x='Positive Review')
plt.title('Distribution of Positive vs Negative Reviews')
plt.xlabel('Positive Review')
plt.ylabel('Count')
plt.show()

# Check for empty or very short reviews
df['Review Length'] = df['Review'].apply(lambda x: len(str(x).split()))
print("\nReview Length Statistics:\n", df['Review Length'].describe())

# Visualize review lengths
sns.histplot(df['Review Length'], bins=50, kde=True)
plt.title('Distribution of Review Lengths')
plt.xlabel('Number of Words')
plt.ylabel('Frequency')
plt.show()

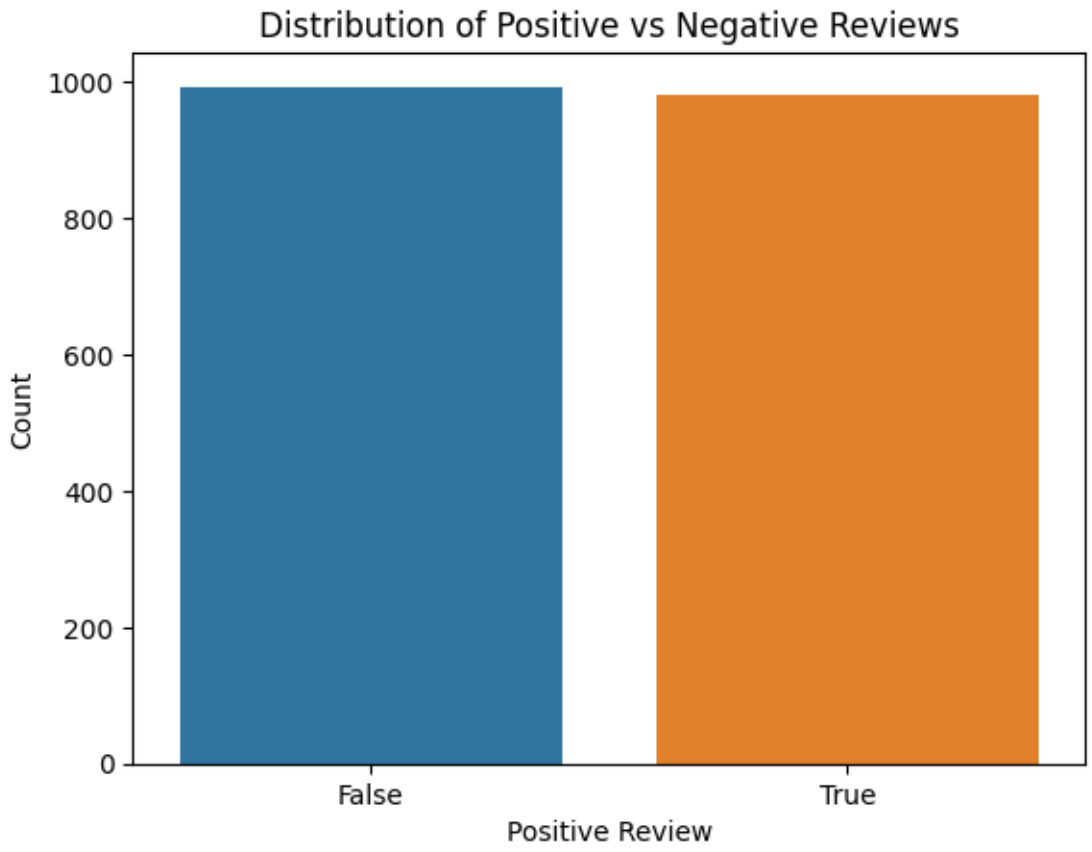
# Example preview of positive and negative reviews
print("\nExamples of Positive Reviews:")
print(df[df['Positive Review'] == True]['Review'].head(3))

print("\nExamples of Negative Reviews:")
print(df[df['Positive Review'] == False]['Review'].head(3))
```

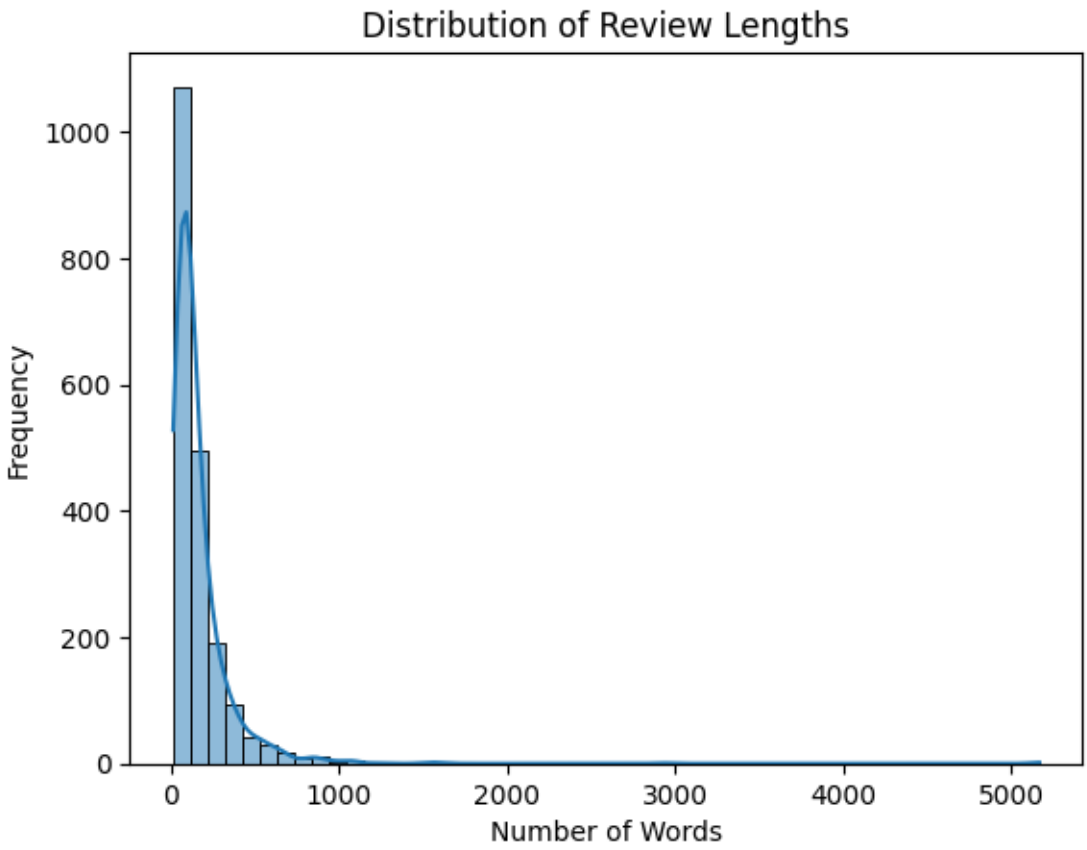
```
Data Types:
 Review             object
Positive Review      bool
dtype: object

Missing Values:
 Review             0
Positive Review    0
dtype: int64

Class Distribution:
 False    993
True     980
Name: Positive Review, dtype: int64
```




```
Review Length Statistics:
 count    1973.000000
mean      160.943234
std       236.136329
min        11.000000
25%        59.000000
50%       103.000000
75%       183.000000
max      5167.000000
Name: Review Length, dtype: float64
```




```
Examples of Positive Reviews:
0    This was perhaps the best of Johannes Steinhof…
1    This very fascinating book is a story written …
2    The four tales in this collection are beautifu…
Name: Review, dtype: object

Examples of Negative Reviews:
3    The book contained more profanity than I expec…
5    I don't know why it won the National Book Awar…
6    The daughter of a prominent Boston doctor is d…
Name: Review, dtype: object
```

1.4 Part 4: Define Your Project Plan

Now that you understand your data, in the markdown cell below, define your plan to implement the remaining phases of the machine learning life cycle (data preparation, modeling, evaluation) to solve your ML problem. Answer the following questions:

- Do you have a new feature list? If so, what are the features that you chose to keep and remove

after inspecting the data?
- Explain different data preparation techniques that you will use to prepare your data for modeling.
- What is your model (or models)?
- Describe your plan to train your model, analyze its performance and then improve the model. That is, describe your model building, validation and selection plan to produce a model that generalizes well to new data.

1. Updated Feature List: After inspecting the data, I have decided to keep the following: - **Feature:** Review (text data) - **Label:** Positive Review (Boolean)

There are no missing values, and both classes (positive and negative reviews) are nearly balanced (993 negative and 980 positive), so no resampling techniques are necessary at this stage.

2. Since this is a Natural Language Processing (NLP) problem, I will apply the following preprocessing steps:

A. Text Cleaning: - Convert all text to lowercase for uniformity. - Remove punctuation, special characters, and numbers using regular expressions.

B. Tokenization: - Split the reviews into individual words.

C. Stopword Removal: - Eliminate common but uninformative words like "the," "and," "is," etc.

D. Lemmatization or Stemming: - Reduce words to their base or root forms to minimize redundancy.

E. Vectorization: - Convert text into numerical format using TF-IDF Vectorization, which highlights important words in each review based on their frequency across the dataset.

3. I will be using a Logistic Regression model to effectively interpret the binary classification using TF-IDF features.

I will also compare my model using the following algorithms:

A. Multinomial Naive Bayes (a common baseline for text classification)

B. Support Vector Machine (SVM) (robust to high-dimensional text data)

C. Random Forest Classifier (to explore non-linear relationships)

4. Here is my outline on how I would like to train, analyze, and improve my model.

Model Building, Validation, and Selection:

A. Train-Test-Split: Split the dataset into training (80%) and testing (20%) sets using train_test_split.

B. Cross-validation: Use 5-fold cross-validation on the training set to evaluate consistency across subsets.

C. Performance Evaluation: Use accuracy, precision, recall, and F1-score as evaluation metrics. - Generate a confusion matrix to inspect false positives and false negatives.

D. Model Improvement: Tune hyperparameters (e.g., regularization strength for logistic regression or C value for SVM) using grid search or randomized search. - Optionally, experiment

with n-gram ranges (e.g., unigrams + bigrams) in the TF-IDF vectorizer to capture phrase-level sentiment.

E. Final Evaluation: Test the final selected model on the hold-out test set to measure real-world generalization.

1.5 Part 5: Implement Your Project Plan

Task: In the code cell below, import additional packages that you have used in this course that you will need to implement your project plan.

[7]:
```
# YOUR CODE HERE

# core packages

import pandas as pd
import numpy as np

# text-related packages
import re
import string
!pip install nltk
import nltk
from nltk.corpus import stopwords
from nltk.stem import WordNetLemmatizer

# sklearn packages for model selection and evaluation
from sklearn.feature_extraction.text import TfidfVectorizer

from sklearn.model_selection import train_test_split, GridSearchCV
from sklearn.linear_model import LogisticRegression
from sklearn.naive_bayes import MultinomialNB
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score, precision_score, recall_score,␣
 ↪f1_score, confusion_matrix, classification_report

# visualization packages
import matplotlib.pyplot as plt
import seaborn as sns

# Download NLTK Resources
nltk.download('stopwords')
nltk.download('punkt')
nltk.download('wordnet')
```

```
Requirement already satisfied: nltk in
/home/ubuntu/.pyenv/versions/3.9.19/lib/python3.9/site-packages (3.9.1)
Requirement already satisfied: click in
```

```
/home/ubuntu/.pyenv/versions/3.9.19/lib/python3.9/site-packages (from nltk)
(8.1.8)
Requirement already satisfied: joblib in
/home/ubuntu/.pyenv/versions/3.9.19/lib/python3.9/site-packages (from nltk)
(1.4.0)
Requirement already satisfied: regex>=2021.8.3 in
/home/ubuntu/.pyenv/versions/3.9.19/lib/python3.9/site-packages (from nltk)
(2024.11.6)
Requirement already satisfied: tqdm in
/home/ubuntu/.pyenv/versions/3.9.19/lib/python3.9/site-packages (from nltk)
(4.67.1)

[notice] A new release of pip is
available: 23.0.1 -> 25.1.1
[notice] To update, run:
pip install --upgrade pip
```

```
[nltk_data] Downloading package stopwords to /home/ubuntu/nltk_data…
[nltk_data]   Unzipping corpora/stopwords.zip.
[nltk_data] Downloading package punkt to /home/ubuntu/nltk_data…
[nltk_data]   Unzipping tokenizers/punkt.zip.
[nltk_data] Downloading package wordnet to /home/ubuntu/nltk_data…
```

[7]: True

Task: Use the rest of this notebook to carry out your project plan.

You will:

1. Prepare your data for your model.
2. Fit your model to the training data and evaluate your model.
3. Improve your model's performance by performing model selection and/or feature selection techniques to find best model for your problem.

Add code cells below and populate the notebook with commentary, code, analyses, results, and figures as you see fit.

[10]:
```python
import re
import string
import nltk
from nltk.corpus import stopwords
from nltk.stem import WordNetLemmatizer

# Download resources (should already be done, but safe to include)
nltk.download('stopwords')
nltk.download('wordnet')

# Initialize stopwords and lemmatizer
stop_words = set(stopwords.words('english'))
lemmatizer = WordNetLemmatizer()
```

```
# Preprocessing function (simple tokenizer using .split())
def preprocess_text(text):
    # Lowercase the text
    text = text.lower()
    # Remove digits
    text = re.sub(r'\d+', '', text)
    # Remove punctuation
    text = re.sub(f"[{re.escape(string.punctuation)}]", "", text)
    # Basic tokenization using split()
    tokens = text.split()
    # Remove stopwords and apply lemmatization
    tokens = [lemmatizer.lemmatize(word) for word in tokens if word not in␣
 ↪stop_words]
    # Reconstruct cleaned sentence
    return " ".join(tokens)

# Apply preprocessing to the 'Review' column
df['Cleaned Review'] = df['Review'].apply(preprocess_text)

# Preview the results
df[['Review', 'Cleaned Review']].head()
```

```
[nltk_data] Downloading package stopwords to /home/ubuntu/nltk_data…
[nltk_data]   Package stopwords is already up-to-date!
[nltk_data] Downloading package wordnet to /home/ubuntu/nltk_data…
[nltk_data]   Package wordnet is already up-to-date!
```

[10]:

	Review	Cleaned Review
0	This was perhaps the best of Johannes Steinhof…	perhaps best johannes steinhoffs book since de…
1	This very fascinating book is a story written …	fascinating book story written form numerous l…
2	The four tales in this collection are beautifu…	four tale collection beautifully composed art …
3	The book contained more profanity than I expec…	book contained profanity expected read book ri…
4	We have now entered a second time of deep conc…	entered second time deep concern science math …

[13]:
```
# TF-IDF Vectorization

# Initialize the vectorizer
vectorizer = TfidfVectorizer(max_features=3000, ngram_range=(1,2))
```

```
# Fit and transform the cleaned reviews
X = vectorizer.fit_transform(df['Cleaned Review'])
y = df['Positive Review']

# Train-Test Split

# Split into training and test sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,␣
↪random_state=42)
```

[14]:
```
# Train and Evaluate Initial Models with Logistic Regression

lr = LogisticRegression()
lr.fit(X_train, y_train) # Fit model

# Predict and evaluate
y_pred = lr.predict(X_test)
print("Classification Report:\n", classification_report(y_test, y_pred))
```

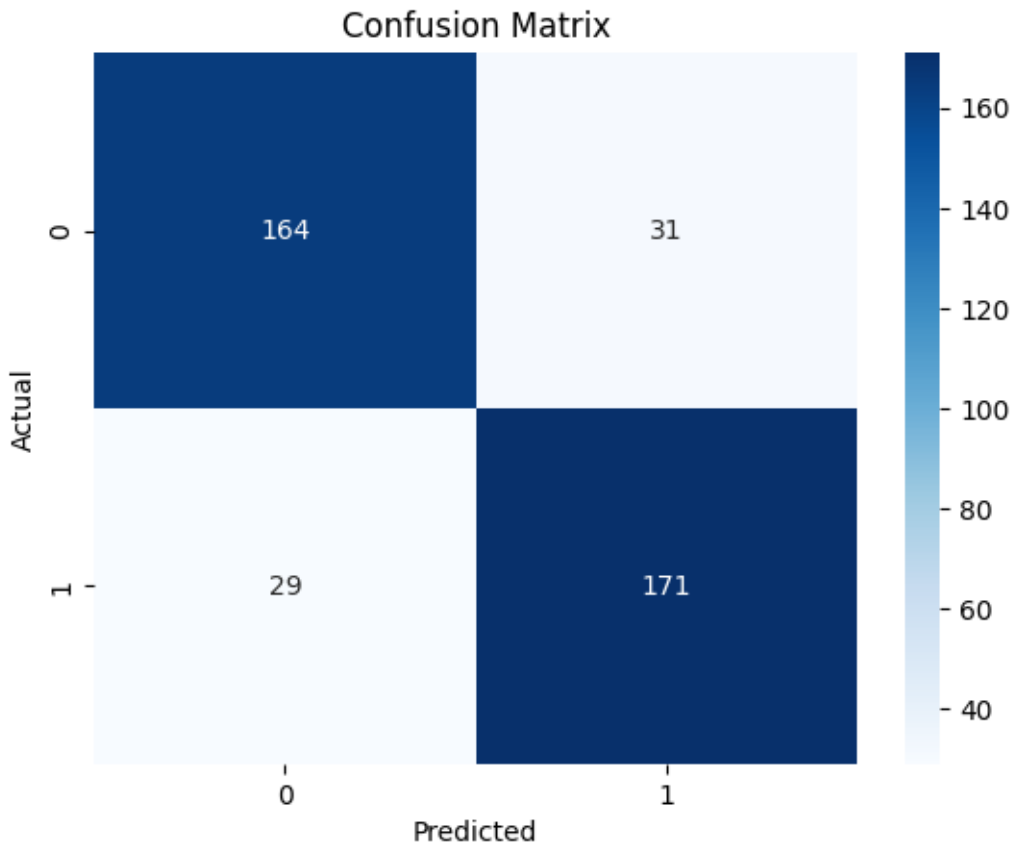
```
Classification Report:
              precision    recall  f1-score   support

       False       0.85      0.84      0.85       195
        True       0.85      0.85      0.85       200

    accuracy                           0.85       395
   macro avg       0.85      0.85      0.85       395
weighted avg       0.85      0.85      0.85       395
```

[15]:
```
# Visualize the confusion matrix to compare and contrast predicted vs. actual␣
↪results

cm = confusion_matrix(y_test, y_pred)
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues')
plt.title("Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()
```



[17]:
```
# Perform grid search for logistic regression
param_grid = {
    'C': [0.01, 0.1, 1, 10],
    'solver': ['liblinear']
}

grid = GridSearchCV(LogisticRegression(), param_grid, cv=5, scoring='f1')
grid.fit(X_train, y_train)

# Print the best model results
y_pred_best = grid.predict(X_test)
print("Best Parameters:", grid.best_params_)
print("Improved Model Report:\n", classification_report(y_test, y_pred_best))
```

```
Best Parameters: {'C': 1, 'solver': 'liblinear'}
Improved Model Report:
              precision    recall  f1-score   support

       False       0.85      0.84      0.85       195
        True       0.85      0.85      0.85       200
```

```
              accuracy                           0.85       395
             macro avg       0.85      0.85      0.85       395
          weighted avg       0.85      0.85      0.85       395
```

[ ]: